

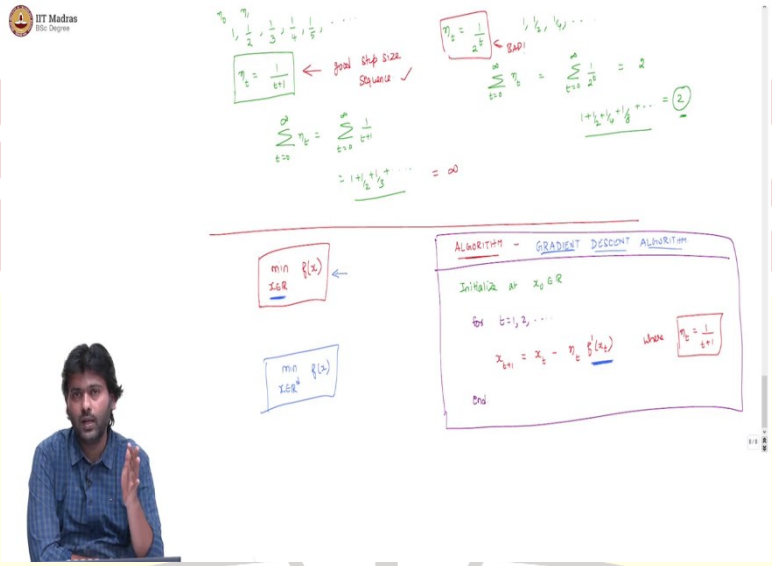
IIT Madras

ONLINE DEGREE

MLF
Assistant Professor Arun Rajkumar
Department of Computer Science & Engineering
Indian Institute of Technology, Madras
Basic Algorithm for Unconstrained optimization: Gradient Descent

So, with that now, we have a very good recipe for making computers solve the minimization problem, we are still in the unconstraint case.

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Which means that we are still trying to understand if we want to minimize x in r f of x. Now, here is the final algorithm. So, let me put down this algorithm, which is basically collecting all the steps that we have said so far. So, initialize at x 0 in r can be anything. Now, for t equals 1 2 . . ., the update rule that we need to use is as follows. So, we do x t + 1 is x t - η t f dash of x t where η in this particular case, we will just set it to 1 over t + 1.

So, this is not the only sequence there might be other sequences with this property, but this is a good enough place to start with. So, this is the algorithm to solve an unconstrained minimization problem. So, it is a very, very simple algorithm, all it needs you to do is you should be able to compute the derivative at a given point, if you can do that and assuming that the function has a derivative and so on.

So, you can actually compute the minimum of that function. Well, let me give you an, give the name of this algorithm and then try to fill in some of the gaps in our understanding so far, there are some questions that you should have in your mind with respect to well, when this algorithm might work, when it might not work and so on. Let me answer those questions after this. But

for the moment, let us first give this algorithm a name and for the moment, the name might seem slightly mysterious, we will also talk about demystifying this name a little bit later.

But the name of this algorithm is called as, this is called as the gradient descent algorithm, it is one of the simplest algorithm in optimization. It is, it is what it is called is a first order algorithm because it uses the first order information, which is the derivative of a function and that is all it needs.

So, but why is it called the gradient descent algorithm? Well, the name gradient can be thought of as a, as a generalization of the derivative to higher dimensions, we did not talk about higher dimensional examples, at least when we came up with this algorithm, we will talk about that as well, next. But it is a general algorithm, it does not work not just for one dimensional data where I mean, one dimensional problems where x is r, it can work for a general problem as well.

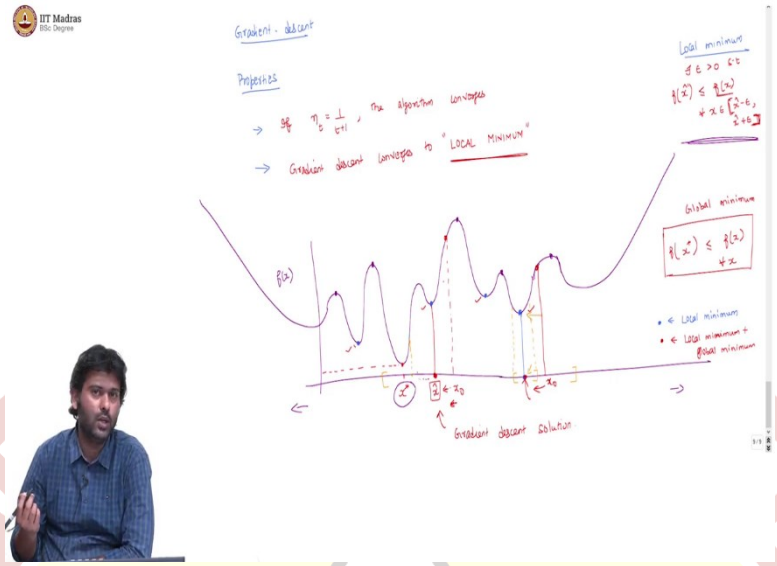
Where you have x in some d dimension, where you want to minimize f of x, where f is now a function that takes as input a d dimensional vector and outputs a real number. So there, instead of the derivative, we will replace it with something called a gradient, which we will come to later.

But for the moment, what we have is an algorithm which is going to work well, when you have a function and you do not have any constraints and you can take the derivative of this function. Now, this is an iterative algorithm, which chooses better and better points as it moves, as t increases and eventually, the hope is that this algorithm will eventually get to optimal solution.

Now, when I say hope is that, we need to understand we should be sure whether this algorithm actually works or not for a general function, so we it is not enough to look at the function that we just looked at, which is x - 5 squared, which is a very simple function. But it should work for any function where derivatives are easy to take.

So now, we would not this course is a very basic course. So, we would not really try to prove a lot of things about when this algorithm will work and when it will not work and so on. I am just going to give you the properties of this algorithm next, to convince you that this is a good algorithm to use in practice. So, that is what we are going to see next.

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So, now what can we say about this algorithm that we have seen, which is the gradient descent algorithm, I am still going to call it the gradient descent. But for the moment, we will still think of this as one dimensional functions. So, properties, properties the main property and one of the most important property is that if for example, if you choose η t as 1 by t + 1, for example, one can show the algorithm converges.

What do I mean, by converges? I mean that there would not be these oscillations that we had, when you did not have a step size. You will indeed converge to an answer. Now, it is not enough that the algorithm actually converges an answer, we also care about converging to good answers.

So, our goal was to find the minimum and now the question is, do we does this algorithm give us what it promised to do? So, we started out with a set out coming to think about an algorithm, which will give us a minimum of a function where there are no constraints and does this algorithm do that? It does that with a caveat.

So, the algorithm gradient descent and this is a very important caveat, converges to what, to what point does this algorithm converge, it converges to what we will call as a local minimum. Now, what we wanted to get is, to get to the minimum of a function. Now, what I am claiming is that this gradient descent algorithm that we just proposed, so very easy algorithm, converges to what is called as a local minimum. Now, what is a local mean? What is the difference between a local minimum and minimum?

Well, it is the easiest to illustrate with an example. Let us say we have a function like this and this is f of x and let us say the function looks like this. Not a sine function, none of that sort. So, I am just making a function, just that it is a wiggly function. So, it has a lot of ups and downs in it. So, now, let us say after this, the function simply goes up. We do not care about its value after the function and let us say we also do not care about this after this point. So, it is, it is only in this point where it is kind of wiggling along. So, the minimum cannot be after the left side here or here. So, it is going to be somewhere here.

Now, if you think of think of what is the minimum of this function? Well, it is clear that the point where the function takes the smallest value is the minimum and it appears here, so this is the minimum value and this is x star. So, x star has the property that f of x star is less than or equal to f of x for all x, the function value at x star is the smallest. So, and that is why it is called a minimum at this point.

So, which is good. So, this is the minimum. Now, when I say local minimum, what do I mean? I mean, these points, this is a local minimum, this is a local minimum, maybe I should use a different colour to illustrate this better, so that I can refer to these, so let me make these blue. This is blue, this is a local minimum. This is a local. So, the blues are local. The red is also a local minimum. We will talk about local minimum in a bit + it is a global minimum.

So, if it is a global minimum, which means, which is just the minimum that we understand intuitively, that f of x star is less than or equal to f of x for all x. So, that is the global minimum. So, you take any point in x, any point x the function value has to be necessarily less than or equal to, greater than or equal to the value of f of x star, that is why it is the global minimum.

The minimum property holds globally, for global x with respect to which x that you are comparing against, so you can take any x and its value will be less than or equal to that value of x. On the other hand, for local minimum, the minimum property, the minimality does not hold, the minimality property does not hold in general, but it only holds locally.

What do I mean by that? Let us, for the purposes of example, let us take, let us consider this point, which is a local minimum, which we claim as a local minimum, what do I mean by local minimum, which means that locally around this point, I should be able to draw an interval in the domain.

So, where within that domain, let us say the function value is lowest at this point. So, within this orange interval, I can take any point and compare the value of the function at that point

with the blue point, well the blue points value will be smaller. But then this does not hold for all intervals.

For example, if I take the interval to be this, now this is no longer a minimum, maybe in this case, in this particular interval it is, maybe I should change the right to here, maybe if I change the interval to be this big interval. Now, this blue point is no longer a local minimum, because there exists some other point in this larger interval, which I can show has a lower value than strictly lower value than the value at this blue point.

Because maybe here is a point. So, x star of course is there, but there are other points also, so maybe this is a point which has smaller value than this point. Now, for something to be characterized as a local minimum, the only necessity is that there should be some interval, so around that point, such that the value of the function at the point is smaller or equal to the value of any point in that interval.

I do not care how small this interval is, as long as there is some interval, I will call this point a local minimum. So, that is the definition of local minimum. So, which means that, so how do I define a local minimum? Well, the formal definition is that, so there exists an epsilon greater than 0, such that f of x hat, let us call this x hat is less than or equal to f of x for all x in x hat - epsilon comma x hat + epsilon, in this interval.

That is what this interval symbol means, I take any point within x hat - epsilon x hat + epsilon, I compare the value of the function with f of x hat, if f of x hat is less than or equal to f of x, then I call this local minimum. Now, of course, if you, I mean pictorially, it just means that these points where these bumps occur, so our work are simply the local minimums.

Equivalently, you can define local maximums as these points, think about what the definition of this local maximum is going to be, it is going to be something very similar to local minimum just that the equality sign, inequality sign is going to get switched. So, why do we care about this.

So, we care about this because for gradient descent, all we can say is that you start with some x 0 arbitrarily, you will only converge to a local minimum, which means that we really want x star which is the global minimum, but unfortunately gradient descent will not necessarily converge to x star.

So, for example, so if I for instance, if I started this was my x 0. If this was my x 0, maybe I will just take a careful x 0 to illustrate this. So, this was my x 0, let us say which means my

functions value is somewhere here. Now, I will find that the direction of movement is to the towards the left, I will keep moving to the left and what will happen is that I will reach this local minimum x hat and I will converge to this x hat.

So, this if I start with this x 0, this is the gradient descent solution. If I start with an x 0 here, let us say I will move this side and then this would be my gradient descent solution for this particular x 0 and so on. So, it is as if, imagine that you are in some part, some point in the hill and then you are sliding down and that you are descending the hill and then at some point the hill again climbs up but then you are stuck at the valley. So, that is the local minima.

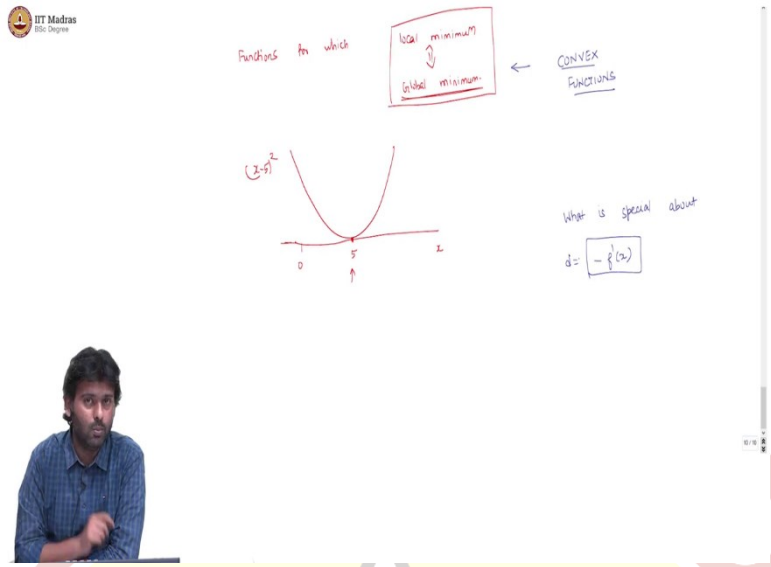
Now, question is and this is an interesting question for you to think about. Now, why does gradient descent get stuck in a local minimum? Now, why cannot it proceed further? What happens? What is so special about this local minimum point that does not allow gradient descent to further proceed.

So, I started with this x 0, this example, I kept moving I got stuck at x hat, I could not move further to the left. So, I get stuck at x hat. So, why does that happen? What property of the algorithm itself forces it to stop at x hat and does not allow it to proceed further? Think about that. What?

So, while I let you think about that, I should also let you know that, well it is very, very hard for any algorithm to get to the global minimum. So, we do not have general purpose algorithms, for any function. To compute the global minimum, that is simply not, it is, it is almost impossible to do that, we can only hope to get to local minimums and if you do not make any further assumptions about the structure of the function itself, well that is the best that you can do.

So, you cannot do better than getting to a local minimum. That is point number one. Point number two is that well, not all is lost in a lot of functions that we really care about in machine learning, getting to a local minimum is actually good enough. So, why do I say that I say that, because in a lot of machine learning applications, you will have functions for which all local minimums are also global minimums.

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So, there are functions for which all local minimum, well local minimum implies global minimum. Well, here is a simple example that we already saw, which was our favourite example that we have kind of been looking at so far, sorry about that, which is our x - 5 squared.

So, here for this particular function, there is only one local minimum, which is 5 and it turns out this 5 is also a global. So, and this is a function for which your gradient descent will indeed converge to a local minimum, but because local minimum also happens to be the global minimum of the function, you also got to the global minimum.

So, which means that there are some special functions for which local minimum is same as global minimum and these functions are those where gradient descent will indeed give you the optimal answer and it turns out that these functions are those that we will use a lot in machine learning, which means that gradient descent will be a very, very powerful tool to understand machine learning ideas.

So, we will talk more about these functions in a later part of the optimization course and let me just give you a name for these functions, these functions are also called as convex functions. We would not talk about this for a while, we will revisit convex functions later, but these are functions for which well, as long as you can find the local minimum, you are also guaranteed to find a global and our gradient descent does a very good job at doing that.

So, with that, I think what we have done now is we have come up with an algorithm, the gradient descent algorithm, which we claim will probably we did not prove it, but then we can one can show that it will probably converge to a local minimum of any function as long as you are able to compute the derivative of this function, that is point number one.

Now, the second thing that we need to understand, there are some questions that should that should naturally occur in our mind, so what is special about the direction that we actually picked. So, what is special about - f dash of x. So, this is the direction that we picked and it seemed to work well for x - 5 squared.

But we did not really argue or even given an intuitive argument as to why this should actually work in general for any function. We will talk about this and more in the coming lectures. But this is a very important question and the reason why it is important to understand why f dash affects or - f dash of x is a good idea is a good direction to pick will also lead us to understanding geometrically what are good directions to move in, when you do have extra conditions or constraints in your problem.

These are things that we will start looking at in the next lectures. For the moment, I want to summarize by saying that we have an algorithm, which is the gradient descent a very powerful algorithm and this algorithm can solve any type of functions and it might, it will give you a local minimum. Thank you.